3.3.1 The cellular basis of cancer and treatment

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the factors that may increase the risk of developing non-communicable diseases	Factors to include heredity, ageing, types of radiation, carcinogen, viruses and air pollution and diseases to include cancers and asthma **AND** to include an evaluation of epidemiological and other evidence to identify correlations.
(b)	the evidence for causal relationships between certain factors and lung disease	Factors to include heredity, ageing, types of radiation, carcinogen, viruses and air pollution **AND** to include an evaluation of epidemiological and other evidence to identify correlations.
(c)	the cellular basis of cancer	To include an outline of cell cycle control and the changes in control which lead to the formation of tumours and metastases.
(d)	how mutations to proto-oncogenes can lead to cancer	To include Ras and Myc proto-oncogenes.
(e)	how mutations to tumour suppressor genes can lead to cancer	To include the p53 gene.

(f)	the evaluation of epidemiological evidence linking potential risk factors with particular forms of cancer	To include smoking and lung cancer, diet and bowel cancer, BRCA1 gene mutations and breast cancer. *M1.3, M1.5, M1.7, M3.1* HSW5, HSW6
(g)	the methods used to detect cancers	To include references to MRI, X-rays, mammography, CT scans, ultrasound, PET scans, biopsies and blood tests.
(h)	the ethical and economic considerations when screening and conducting genetic tests for cancer	To include evaluation of screening for particular cancers e.g. the potential harm, accuracy and cost of the screening procedure **AND** discussion of the ethics of genetic tests e.g. for BRCA and HNPCC genes. HSW9, HSW10
(i)	the methods used to treat patients with cancer.	To include surgery, chemotherapy, radiotherapy, immunotherapy (monoclonal antibodies), complementary therapies and hormone-related treatment.